# MineTrace - Critical Mineral Supply-Chain Risk Assessment

**REPORT ID: REP-9A24723FB616 | ANALYSIS ID: DOWNLOAD-1 | TEMPLATE VERSION: 1.0**

*Generated At: 2026-10-27T10:24:10 UTC | Original Analysis Timestamp: 2023-06-01T12:00:00 UTC*

## 1. Executive Summary

**CLASSIFICATION: RED (ANOMALY SCORE: 60.0/100) - COMPLETE**
Flag Status: FLAGGED | Sanctions Screening: NO_IDENTIFIED_MATCH | Evaluated Rules: 5 | Insufficient Rules: 0

| | | | |
|---|---|---|---|
| **Mineral / Commodity:** | Lithium carbonate (HS 283691) | **Reporter Country:** | CHL |
| **Partner Country:** | USA | **Trade Period:** | 2023 |
| **Score Completeness:** | COMPLETE | **Sanctions Screening:** | NO_IDENTIFIED_MATCH |

## 2. Risk Rule Evaluation Details

| Rule ID | Rule Name | Weight | Outcome | Threshold | Calculated |
|---|---|---|---|---|---|
| R1 | Mirror Trade Gap | 30 pts | FLAG | > 30.0% | 45.0% |

[R1] Mirror Trade Gap (Weight: 30 pts) - Status: FLAG
Formula: abs(export_value - import_value) / max(export_value, import_value) * 100 | Threshold: > 30.0% | Calculated: 45.0%
Inputs: {'export_value': 1200000.0, 'import_value': 660000.0}
Explanation: Export-import mirror trade gap 45.0% exceeds threshold.

## 3. Classification & Sufficiency Guardrails

| | | | |
|---|---|---|---|
| **Saved Anomaly Score:** | 60.0 / 100.0 | **Assigned Tier:** | RED |
| **Flag Status:** | FLAGGED | **Policy Version:** | 1.0 |

Thresholds Applied: GREEN (0-19), AMBER (20-49), RED (50-100), INSUFFICIENT_DATA (PARTIAL/UNAVAILABLE)
Classification Rationale: Elevated anomaly score exceeding RED prototype threshold (>= 50.0).

## 4. Sanctions Screening (OFAC SDN)

**SANCTIONS STATUS: NO_IDENTIFIED_MATCH**
Registry Reference: OFAC Specially Designated Nationals (SDN) List Limitations:
SANCTIONS SCREENING LIMITATIONS: Screening is performed against public US Treasury OFAC Specially Designated Nationals (SDN) registries. Country-level or commodity-level screening does not substitute for corporate beneficial ownership checks (e.g., 50 Percent Rule) or end-user certification.

## 5. Source Evidence & Audit Trail

| | | | |
|---|---|---|---|
| **Source Dataset:** | UN Comtrade | **Source Record ID:** | TRADE-1 |
| **Retrieval Timestamp:** | 2023-05-01T10:00:00 | **Evidence Snapshot ID:** | SNAP-DOWNLOAD-1 |
| **Validation Status:** | VALID | **Source Endpoint / URL:** | https://comtradeapi.un.org/public/v1/pre |

## 6. Validated AI Explanation

*Status: NOT_REQUESTED. No pre-validated AI explanation was attached to this transaction record.*

## 7. Methodology, Limitations & Legal Notice

**NOTICE & DISCLAIMER: This document is an automated evidence report generated by the MineTrace critical mineral supply-chain risk intelligence system. Numerical anomaly scores, rule evaluations, and risk classifications (GREEN, AMBER, RED, INSUFFICIENT_DATA) reflect deterministic statistical and relational anomalies compared against historical baselines and official trade registries. They DO NOT constitute a legal finding of fraud, illicit trade, customs violation, or sanctions evasion. A GREEN classification does not certify a supply chain as compliant, ethical, or risk-free. Missing or incomplete data must never be interpreted as low risk.**

METHODOLOGY & LIMITATIONS:

1. Anomaly Score: A deterministic weighted score (0-100) aggregating multi-rule trade indicators.
2. Scope of Detection: Risk rules evaluate reported trade flows (Comtrade) and public SDN lists (OFAC). They do not inspect physical shipping containers or corporate beneficial ownership trees.
3. Data Sufficiency: Where required historical baselines or partner trade reports are unavailable, MineTrace strictly enforces sufficiency guardrails and assigns INSUFFICIENT_DATA rather than assuming compliance.
4. Sanctions Screening: Evaluates exact and normalized name matches against US Treasury OFAC records. A status of NO_IDENTIFIED_MATCH does not verify compliance with non-US regimes or unlisted intermediaries.
5. Prototype Status: Thresholds (0-19 GREEN, 20-49 AMBER, 50-100 RED) are heuristic operational baselines designed for screening and prioritization, not statistically validated probabilities of wrongdoing.